APEX RETAIL INTELLIGENCE

END-TO-END DATA ENGINEERING AND ANALYTICS PIPELINE

Detailed Project Documentation

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1. Project Information

Item	Details
Project Name	APEX Retail Data Engineering Pipeline
Project Type	End-to-End Data Engineering and Analytics
Platform	Databricks
Processing Engine	Apache Spark / PySpark
Programming Language	Python
Query Language	SQL
Storage Formats	CSV, Parquet, Delta Lake
Data Catalog	Unity Catalog
Catalog Name	apex_retail
Gold Schema	gold
Primary Output	Gold Star Schema
Analytical Output	KPI, Sales, Customer, Product, RFM and Cohort Analytics

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2. Executive Summary

The **APEX Retail Data Engineering Pipeline** is an end-to-end data engineering project designed to process, clean, validate, transform, model, and analyze retail data using modern data engineering technologies.

The project demonstrates how raw retail data can be transformed into reliable and business-ready analytical datasets.

The pipeline processes three major business domains:

1. Customer data
2. Product data
3. Sales transaction data

The project starts with raw CSV data and performs ingestion, validation, cleaning, transformation, incremental processing, Delta Lake operations, Delta MERGE, Slowly Changing Dimension processing, surrogate key generation, dimensional modeling, and analytical reporting.

The final data is organized into a **Gold-layer Star Schema** consisting of:

- dim_customer
- dim_product
- dim_promotion
- dim_date
- fact_sales

The Gold tables are registered under the Unity Catalog namespace:

apex_retail.gold

The project also implements business analytics including:

- Sales KPI reporting
- Revenue analysis
- Customer analysis
- Product analysis
- Promotion analysis
- RFM customer segmentation
- Cohort retention analysis

A comprehensive validation framework was implemented to verify:

- Data quality
- NULL values
- Duplicate records
- Transaction uniqueness
- Foreign-key integrity

- Sales reconciliation
- Customer reconciliation
- Product reconciliation
- Gold table structure

The final pipeline audit confirms the successful completion of the data engineering workflow.

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3. Introduction

Retail businesses generate large amounts of data every day through customer interactions, product sales, promotions, stores, and transactions.

Although raw data contains valuable business information, it is usually not immediately suitable for analytical reporting.

Raw datasets may contain:

- Missing values
- Duplicate records
- Incorrect data types
- Invalid dates
- Invalid identifiers
- Inconsistent values
- Unmatched references
- Historical changes

Using such data directly for business reporting can result in inaccurate KPIs and incorrect business decisions.

Therefore, a reliable data engineering pipeline is required to convert raw data into clean, consistent, validated, and analytics-ready information.

The APEX Retail project addresses this requirement by implementing an end-to-end data pipeline using Databricks and PySpark.

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4. Business Problem Statement

The objective of the project is to build a reliable data processing system for a retail organization.

The retail organization needs to answer questions such as:

- How much revenue was generated?
- Which products generate the highest revenue?
- Which customers are most valuable?
- Which promotions perform best?
- How many transactions were completed?
- What is the average order value?
- Which customers are loyal?
- Which customers may be at risk of becoming inactive?
- How well are customers retained over time?
- Are sales transactions correctly linked to customers and products?

Raw source data cannot reliably answer these questions without proper data engineering.

Therefore, the project creates a structured pipeline that transforms raw retail data into a trusted analytical platform.

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5. Project Objectives

5.1 Primary Objectives

The primary objectives are:

1. Build a complete retail data engineering pipeline.
2. Ingest CSV source data using PySpark.
3. Perform schema and data-type validation.
4. Clean customer, product, and sales datasets.
5. Implement data-quality rules.
6. Convert raw data into efficient columnar formats.
7. Implement incremental processing.
8. Use Delta Lake for reliable data storage.
9. Implement Delta MERGE.

10. Implement SCD Type 1 and Type 2 concepts.
11. Generate surrogate keys.
12. Build a Star Schema.
13. Register Gold tables in Unity Catalog.
14. Generate business KPIs.
15. Perform customer and product analytics.
16. Perform RFM analysis.
17. Perform cohort retention analysis.
18. Perform reconciliation.
19. Perform final data-quality validation.
20. Produce a final pipeline status.

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6. Scope of the Project

The project covers the following areas:

Data Ingestion

- CSV ingestion
- Raw data processing
- File organization

Data Processing

- PySpark transformations
- Data cleaning
- Type conversion
- NULL handling
- Duplicate handling

Data Storage

- Parquet
- Delta Lake

- Unity Catalog

Data Modeling

- Dimension tables
- Fact table
- Star Schema
- Surrogate keys

Data Management

- Incremental processing
- Delta MERGE
- SCD Type 1
- SCD Type 2

Analytics

- Sales KPIs
- Customer analytics
- Product analytics
- Promotion analytics
- RFM analysis
- Cohort retention

Data Quality

- NULL checks
- Duplicate checks
- Referential integrity
- Transaction validation
- Reconciliation
- Final audit

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7. Technology Stack

7.1 Databricks

Databricks was used as the main development and execution environment.

It provides:

- Notebook-based development
- Apache Spark integration
- Delta Lake support
- SQL support
- Unity Catalog integration
- Scalable data processing

7.2 PySpark

PySpark was used for distributed data processing.

Major operations include:

- Reading data
- Filtering
- Selecting columns
- Joining datasets
- Aggregating data
- Creating calculated columns
- Window functions
- Data validation
- Writing Delta tables

7.3 Python

Python was used for:

- Pipeline logic
- Data transformations

- Validation logic
- Analytics
- Audit reporting
- Control flow

7.4 SQL

SQL was used for:

- Table inspection
- Data validation
- Analytical queries
- Aggregations
- Table management
- Unity Catalog verification

7.5 Parquet

Parquet is a columnar storage format used during data processing.

Advantages include:

- Efficient storage
- Compression
- Column pruning
- Faster analytical queries
- Better Spark performance

7.6 Delta Lake

Delta Lake was used for reliable table storage and data processing.

Important capabilities include:

- ACID transactions

- MERGE
- Schema management
- Version history
- Reliable updates
- Incremental processing

7.7 Unity Catalog

Unity Catalog was used to manage the Gold-layer tables.

The project uses:

Catalog: apex_retail

Schema : gold

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8. Overall System Architecture

The project follows a structured data engineering architecture.

```
┌──────────────────────────┐
│   RAW CSV DATA     │
└────────────┬─────────────┘
      │
      ▼
┌──────────────────────────┐
│ RAW / INBOUND      │
│ PROCESSING        │
└────────────┬─────────────┘
      │
      ▼
┌──────────────────────────┐
│ SCHEMA & DATA      │
```

QUALITY VALIDATION

▼

PARQUET CONVERSION

▼

DATA CLEANING
& TRANSFORMATION

▼

INCREMENTAL
PROCESSING

▼

DELTA LAKE
+ MERGE

▼

```
┌──────────────────────────┐
│ SCD TYPE 1 / TYPE 2      │
│ SURROGATE KEYS           │
└────────────┬─────────────┘
     │
     ▼
┌──────────────────────────┐
│ GOLD STAR SCHEMA         │
└────────────┬─────────────┘
     │
     ▼
┌──────────────────────────┐
│ UNITY CATALOG            │
└────────────┬─────────────┘
     │
     ▼
┌──────────────────────────┐
│ ANALYTICS & KPIs         │
└────────────┬─────────────┘
     │
     ▼
┌──────────────────────────┐
│ RECONCILIATION &         │
│ FINAL AUDIT              │
└──────────────────────────┘
```

---

9. Data Flow

The complete data flow is:

```
Customer CSV ──────┐
            │
Product CSV ───────┼──► Raw Processing
            │
Sales CSV ─────────┘
              │
              ▼
          Data Cleaning
              │
              ▼
        Data Quality Checks
              │
              ▼
      Incremental Processing
              │
              ▼
           Delta Data
              │
              ▼
          MERGE / SCD
              │
              ▼
         Gold Dimensions
              │
              ▼
           Fact Sales
```

|

▼

Star Schema

|

▼

KPI Analytics

|

▼

RFM / Cohort Analysis

|

▼

Reconciliation / Audit

---

10. Source Data Description

10.1 Customer Dataset

The customer dataset contains demographic and customer-related attributes.

Important fields include:

Column	Description
customer_id	Unique customer identifier
age	Customer age
gender	Customer gender attribute
income_bracket	Income category
loyalty_program	Loyalty membership indicator
membership_years	Years in loyalty program
churned	Customer churn indicator

Column	Description
marital_status	Marital status
number_of_children	Number of children
education_level	Education information
occupation	Customer occupation
customer_zip_code	Customer ZIP code
customer_city	Customer city
customer_state	Customer state

---

11. Product Dataset

The product dataset acts as the master reference for products.

The product dataset is used for:

- Product identification
- Product validation
- Product analytics
- Sales-product relationships
- Product dimension creation

The product master is particularly important because sales transactions reference products using a product identifier.

---

12. Sales Dataset

The sales dataset contains transaction-level records.

Important attributes include:

Attribute	Purpose
transaction_id	Unique transaction identifier

Attribute	**Purpose**
transaction_date	Transaction date
customer_id	Customer reference
product_id	Product reference
quantity	Number of units sold
unit_price	Price per unit
discount	Applied discount
payment_method	Payment type
store_location	Store information
promotion_id	Promotion reference
total_sales	Transaction revenue

The sales dataset is the primary source for fact_sales.

---

13. Raw / Inbound Processing

The first stage of the pipeline is raw/inbound processing.

The objective is to bring source data into the processing environment without applying excessive business transformations.

The process includes:

1. Read CSV files.
2. Inspect schemas.
3. Check column names.
4. Validate data types.
5. Identify missing values.
6. Identify duplicates.
7. Convert data to Parquet.
8. Prepare data for downstream processing.

---

14. Schema Inspection

Schema inspection ensures that incoming data matches expected structures.

For example:

customer_id → string

age → integer

quantity → numeric

transaction_date → date

Schema inspection helps detect problems early in the pipeline.

---

15. Data Cleaning

Data cleaning is performed before building analytical datasets.

The main cleaning categories are:

Customer

- Remove or handle invalid IDs
- Check duplicates
- Validate attributes
- Standardize types

Product

- Validate product identifiers
- Check duplicates
- Validate attributes

Sales

- Validate transaction identifiers
- Validate dates
- Validate quantities
- Validate revenue

- Validate references

---

16. NULL Validation

NULL checks are performed on important columns.

Critical business keys include:

customer_id

product_id

transaction_id

Critical transaction attributes include:

transaction_date

quantity

total_sales

The final pipeline confirms that critical fact-table surrogate keys are not NULL.

---

17. Duplicate Validation

Duplicates can cause incorrect reporting.

For example, duplicate transaction IDs can result in:

- Double-counted revenue
- Incorrect transaction counts
- Incorrect customer frequency
- Incorrect KPI values

Therefore, the project explicitly checks transaction uniqueness.

Final result:

NULL Transaction IDs : 0

Duplicate Transaction IDs : 0

---

18. Incremental Processing

Incremental processing allows the pipeline to process new or changed records without rebuilding everything unnecessarily.

The concept is:

```
Existing Dataset
       +
Incoming Dataset
       │
       ▼
Compare Records
       │
 ┌─────────┴─────────┐
 │          │
Changed    New
 │          │
 ▼          ▼
Update    Insert
```

This approach is important for scalable production pipelines.

---

19. Delta Lake Architecture

Delta Lake combines the benefits of Parquet with transactional reliability.

The project uses Delta for processed and Gold datasets.

A Delta table consists conceptually of:

```
Delta Table
│
├── Parquet Data Files
│
└── _delta_log
```

The transaction log maintains information about changes to the table.

---

20. Delta MERGE

The MERGE operation is used for incremental updates.

Conceptually:

```
Source
  |
  ▼
Match Target?
  |
  ├── YES → UPDATE
  |
  └── NO  → INSERT
```

This supports efficient data synchronization.

---

21. Slowly Changing Dimensions

Dimension data can change over time.

For example, a customer's city or occupation may change.

The project demonstrates SCD techniques.

21.1 SCD Type 1

Type 1 replaces the previous value.

Example:

Before:

City = Mumbai

After:

City = Pune

Only the latest value remains.

21.2 SCD Type 2

Type 2 preserves historical values.

Example:

Customer ID: C101

Version 1:

City = Mumbai

Valid From = 2025-01-01

Valid To = 2026-01-15

Current = False

Version 2:

City = Pune

Valid From = 2026-01-16

Valid To = NULL

Current = True

This allows analysts to understand historical customer states.

---

22. Surrogate Key Generation

Surrogate keys are artificial keys generated for analytical dimensions.

The project uses:

customer_sk

product_sk

promotion_sk

date_sk

Benefits include:

- Stable dimension references
- Better Star Schema design
- Support for historical dimension versions
- Separation between source keys and warehouse keys

---

23. Gold Layer Design

The Gold layer is the final business-ready layer.

It contains:

apex_retail.gold

The main tables are:

dim_customer

dim_product

dim_promotion

dim_date

fact_sales

gold_kpi

sales_gold

---

24. Star Schema Design

The Gold layer follows dimensional modeling principles.

```
            DIM_CUSTOMER
                |
                |
                ▼
DIM_PRODUCT ──── FACT_SALES ──── DIM_PROMOTION
```

|
|
▼
DIM_DATE

The fact table is located at the center and dimensions provide descriptive context.

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25. Dimension Table Details

25.1 Customer Dimension

The customer dimension stores descriptive customer information.

Primary analytical identifier:

customer_sk

Source business identifier:

customer_id

25.2 Product Dimension

The product dimension stores product master information.

Primary analytical identifier:

product_sk

Source business identifier:

product_id

25.3 Promotion Dimension

The promotion dimension stores promotional information.

Primary analytical identifier:

promotion_sk

25.4 Date Dimension

The date dimension supports time-based analysis.

It allows reporting by:

- Day
- Week
- Month
- Quarter
- Year

Primary identifier:

date_sk

---

26. Fact Sales Design

fact_sales is the central transactional table.

It contains:

Foreign Keys

customer_sk

product_sk

promotion_sk

date_sk

Business Key

transaction_id

Measures

Examples include:

quantity

total_sales

The fact table connects transactional measures with descriptive dimensions.

---

27. Referential Integrity

Referential integrity ensures that fact records correctly reference dimension records.

The project validates:

fact_sales.customer_sk → dim_customer

fact_sales.product_sk → dim_product

fact_sales.promotion_sk → dim_promotion

fact_sales.date_sk → dim_date

The final validation result is:

NULL Customer SKs : 0

NULL Product SKs : 0

NULL Promotion SKs : 0

NULL Date SKs : 0

---

28. Unknown Dimension Strategy

In real-world data warehouses, not every transaction will always have a matching dimension record.

The project handles such cases using an Unknown Dimension key.

-1 = Unknown

For example:

Unknown Product → product_sk = -1

Unknown Date → date_sk = -1

This approach prevents NULL foreign keys and preserves fact records.

---

29. Invalid Transaction Dates

Some transactions contained:

transaction_date = "-"

This is not a valid date.

The pipeline avoids creating a fake date.

Instead, the record receives:

date_sk = -1

This preserves the transaction and makes the missing/invalid date explicit.

---

30. Unity Catalog Registration

The Gold tables are registered using the namespace:

apex_retail.gold

The table structure can be verified with:

SHOW TABLES IN apex_retail.gold;

This provides a centralized location for analytical datasets.

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31. KPI Reporting

The project produces business-level KPIs.

Major metrics include:

Total Revenue

Total sales generated by all transactions.

Total Transactions

Number of unique transactions.

Total Quantity

Total number of units sold.

Average Order Value

Average revenue generated per transaction.

Conceptually:

Average Order Value =

Total Revenue / Total Transactions

---

32. Sales Analytics

Sales analytics provides insight into business performance.

Analysis includes:

- Revenue trends
- Monthly revenue
- Store-level revenue
- Product revenue
- Promotion revenue
- Quantity analysis
- Payment method analysis

---

33. Customer Analytics

Customer analytics focuses on customer purchasing behavior.

Metrics include:

- Customer transaction count
- Customer revenue
- Purchase frequency
- Quantity purchased
- Customer contribution

This helps identify valuable customers.

---

34. Product Analytics

Product analysis includes:

- Total units sold
- Product revenue
- Product ranking
- Top-selling products
- Product contribution

This helps identify products that drive revenue.

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35. Promotion Analytics

Promotion analysis evaluates the impact of promotions.

Metrics include:

- Sales by promotion
- Revenue by promotion
- Transaction count by promotion
- Customer activity by promotion

---

36. RFM Analysis

RFM is a customer segmentation methodology.

It consists of:

R = Recency

F = Frequency

M = Monetary

36.1 Recency

Measures the number of days since the customer's most recent transaction.

A more recent customer generally receives a better recency score.

36.2 Frequency

Measures the number of transactions made by the customer.

Higher frequency indicates stronger customer engagement.

36.3 Monetary

Measures the total amount spent by the customer.

Higher monetary value indicates a higher-value customer.

36.4 RFM Segmentation

RFM scores can be used to identify groups such as:

Champions

Loyal Customers

Potential Loyalists

Recent Customers

At Risk

Needs Attention

Low Value

The segmentation supports targeted marketing and customer-retention strategies.

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37. Cohort Analysis

Cohort analysis groups customers based on their first purchase period.

For example:

January Cohort

February Cohort

March Cohort

April Cohort

Each cohort is then tracked across subsequent periods.

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38. Cohort Retention Matrix

The project generates a retention matrix using:

cohort_month

cohort_month_number

retention_percentage

The matrix helps answer:

- How many customers return after their first purchase?
- Which cohorts have the strongest retention?
- How does retention change over time?

---

39. Data Reconciliation

Reconciliation verifies that important business values remain consistent throughout the pipeline.

The project compares:

Record Count

Source transaction count versus Gold transaction count.

Quantity

Source quantity versus fact quantity.

Revenue

Source revenue versus fact revenue.

Final result:

Sales Reconciliation : PASS

---

40. Customer Reconciliation

Customer records are compared across the processing stages.

The purpose is to verify that:

- Customers are not unexpectedly lost.
- Customer identifiers remain consistent.
- Customer transformations are correct.

Final result:

Customer Reconciliation : PASS

---

41. Product Reconciliation

Product reconciliation compares the product master against products referenced by sales.

Initially, unmatched product references were identified.

The pipeline then handled unmatched products through the Unknown Product dimension strategy.

Final result:

Product Reconciliation : PASS

---

42. Final Data Quality Report

The final pipeline validates all major quality dimensions.

Customer Data Quality : PASS

Product Data Quality : PASS

Sales Data Quality : PASS

Star Schema Foreign Keys : PASS

Transaction Validation : PASS

Sales Reconciliation : PASS

Gold Table Structure : PASS

---

43. Final Pipeline Audit

The final audit acts as the final acceptance test.

The audit confirms that the pipeline has successfully completed its required stages.

Expected result:

==

APEX RETAIL FINAL PIPELINE AUDIT

==

Customer Data Quality : PASS

Product Data Quality : PASS

Sales Data Quality : PASS

Star Schema Foreign Keys : PASS

Transaction Validation : PASS

Sales Reconciliation : PASS

Gold Table Structure : PASS

==

FINAL PIPELINE STATUS: PASS

APEX RETAIL PIPELINE COMPLETED SUCCESSFULLY

==

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44. Error Handling and Debugging

During development, several technical issues were encountered and resolved.

44.1 Schema Not Found

An error occurred because the expected schema was not available.

The issue was resolved by checking the actual catalog and schema structure.

The correct namespace became:

apex_retail.gold

44.2 Undefined Spark Window

A NameError occurred because the Spark Window function had not been imported.

The issue was resolved by importing the required Spark SQL functions.

44.3 Undefined DataFrames

Errors such as:

NameError: name 'customer_silver' is not defined

occurred when cells were executed out of sequence.

The issue was resolved by:

- Creating DataFrames before using them.
- Running notebook cells in dependency order.
- Re-executing required transformation cells.

44.4 Invalid Date Column

An unresolved-column error occurred when a column was referenced as:

date

while the actual expression was:

to_date(transaction_date)

The issue was resolved by explicitly creating or referencing the correct date column.

44.5 Invalid Date Values

The value:

-

could not be converted into a valid date.

Unknown Date handling was implemented using:

date_sk = -1

44.6 Missing Gold Tables

The final audit initially reported:

dim_customer

dim_promotion

as missing.

The issue was identified as a Unity Catalog registration issue rather than a data-processing failure.

After registering the required tables, the Gold structure validation passed.

45. Performance Considerations

The project uses several techniques to improve Spark processing efficiency.

Parquet

Reduces unnecessary data scanning.

Delta

Provides efficient transactional storage.

Incremental Processing

Avoids unnecessary full refreshes.

Column Selection

Only required columns should be selected where possible.

Filtering

Filters should be applied as early as practical.

Aggregations

Aggregations are performed using Spark distributed processing.

For larger production datasets, further optimizations could include:

- Partitioning
- Z-Ordering
- Broadcast joins
- Adaptive Query Execution
- Optimized file sizes
- Databricks Workflows

---

46. Data Governance

Unity Catalog provides a centralized governance layer.

Potential governance capabilities include:

- Table ownership

- Access control
- Permissions
- Metadata management
- Data discovery
- Data lineage

The Gold schema acts as a controlled business-ready layer.

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47. Security

Security is an important consideration in a production data pipeline.

The following should never be committed to GitHub:

Passwords

API Keys

Access Tokens

Database Credentials

Secret Keys

Connection Strings

Secrets should be stored using secure mechanisms such as:

- Databricks Secret Scopes
- Environment variables
- Secure configuration systems

---

48. Project Structure

The recommended project structure is:

APEX-Retail-Intelligence /

│

├── README.md

│

```
├── notebooks/
│   └── apex _retail_intelligence_.py
│   └── Updated apex_retail_intelligence.py
│
├── docs/
│   └── APEX_Retail_Detailed_Documentation.pdf
│
├── screenshots/
│
├── data/
│       ├── customer_sample.csv
│       ├── product_sample.csv
│       └── sales_sample.csv
```

---

49. Execution Steps

The complete notebook should be executed in the following sequence.

Step 1 — Load Data

Read customer, product, and sales datasets.

Step 2 — Inspect Data

Check:

- Schema
- Columns
- Data types
- Record counts

Step 3 — Validate Raw Data

Perform:

- NULL checks

- Duplicate checks
- Key checks

Step 4 — Clean Data

Clean and standardize the datasets.

Step 5 — Convert to Parquet

Store processed source data using Parquet.

Step 6 — Incremental Processing

Identify new and changed records.

Step 7 — Delta Processing

Write data into Delta format.

Step 8 — MERGE

Apply updates and inserts.

Step 9 — SCD

Apply Type 1 and Type 2 logic.

Step 10 — Generate Surrogate Keys

Create dimension surrogate keys.

Step 11 — Build Dimensions

Create:

dim_customer

dim_product

dim_promotion

dim_date

Step 12 — Build Fact

Create:

fact_sales

Step 13 — Register Tables

Register tables under:

apex_retail.gold

Step 14 — Generate KPIs

Create sales and business metrics.

Step 15 — RFM Analysis

Segment customers.

Step 16 — Cohort Analysis

Generate retention matrix.

Step 17 — Reconciliation

Compare source and target measures.

Step 18 — Final Audit

Run final validation.

---

50. Testing Strategy

The pipeline uses validation checks at multiple stages.

Unit-Level Checks

Examples:

- Column existence
- Data types
- NULL checks
- Duplicate checks

Transformation Checks

Examples:

- Join correctness
- Calculated fields
- Aggregation correctness

Integration Checks

Examples:

- Fact-dimension relationships
- MERGE results
- SCD behavior

End-to-End Checks

Examples:

- Record reconciliation
- Quantity reconciliation
- Revenue reconciliation
- Final Gold structure

---

51. Acceptance Criteria

The project can be considered successfully completed when:

Raw Processing

- CSV files are successfully processed.
- Raw data is organized.
- Parquet conversion works.

Data Quality

- NULL checks are implemented.
- Duplicate checks are implemented.
- Invalid records are handled.

Incremental Processing

- New and changed records are identified.
- Incremental logic works correctly.

Delta

- Delta tables are created.
- MERGE works correctly.

SCD

- Type 1 logic is implemented.
- Type 2 logic is implemented.

Gold

- Star Schema is created.
- Dimensions are available.
- Fact table is available.
- Surrogate keys are generated.

Unity Catalog

- Gold tables are registered.
- Tables can be queried using the catalog namespace.

Analytics

- KPIs are generated.
- RFM analysis is available.
- Cohort retention is available.

Validation

- Reconciliation passes.
- Foreign-key validation passes.
- Transaction validation passes.
- Final pipeline status is PASS.

---

52. Project Results

The final project achieved the following:

Raw / Inbound Processing : COMPLETED

CSV Processing : COMPLETED

Parquet Processing : COMPLETED

Customer Cleaning : COMPLETED

Product Cleaning : COMPLETED

Sales Cleaning : COMPLETED

NULL Validation : PASS

Duplicate Validation : PASS

Incremental Processing : COMPLETED

Delta Lake : COMPLETED

Delta MERGE : COMPLETED

SCD Type 1 : COMPLETED

SCD Type 2 : COMPLETED

Surrogate Key Generation: COMPLETED

Gold Star Schema : COMPLETED

Unity Catalog : COMPLETED

Sales Analytics : COMPLETED

Customer Analytics : COMPLETED

Product Analytics : COMPLETED

KPI Reporting : COMPLETED

RFM Analysis : COMPLETED

Cohort Retention : COMPLETED

Customer Reconciliation : PASS

Product Reconciliation : PASS

Sales Reconciliation : PASS

Foreign-Key Validation : PASS

Transaction Validation : PASS

Final Data Quality : PASS

Final Pipeline Audit : PASS

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53. Key Learning Outcomes

The project provides practical understanding of modern data engineering.

Technical Learning

The project demonstrates practical experience with:

- PySpark
- Spark SQL
- Databricks
- Delta Lake
- Parquet
- Unity Catalog
- DataFrames
- Data transformations
- Joins
- Aggregations
- Window functions
- MERGE

Data Engineering Learning

The project demonstrates:

- Data ingestion
- Data cleaning
- Data validation

- Incremental processing
- SCD
- Dimensional modeling
- Star Schema
- Data reconciliation
- Pipeline auditing

Analytics Learning

The project demonstrates:

- KPI development
- Customer segmentation
- RFM
- Cohort analysis
- Retention analysis
- Sales analytics

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54. Future Enhancements

Although the current pipeline successfully meets the core project objectives, several production-level improvements can be added.

54.1 Dedicated Bronze Layer

A dedicated Bronze Delta layer can be introduced between Raw and Silver.

Raw

↓

Bronze

↓

Silver

↓

Gold

This would provide stronger separation between raw ingestion and cleaned data.

54.2 Databricks Workflows

The notebook can be converted into scheduled jobs using Databricks Workflows.

54.3 Automated Data Quality

Data-quality checks can be automated using frameworks such as:

- Great Expectations
- Custom DQ frameworks
- Databricks expectations

54.4 BI Dashboard

The Gold layer can be connected to:

- Power BI
- Tableau
- Databricks SQL dashboards

54.5 CI/CD

The project can be integrated with GitHub Actions or another CI/CD platform for automated testing and deployment.

54.6 Advanced Analytics

Future versions could include:

- Customer lifetime value
- Churn prediction
- Demand forecasting
- Product recommendation

- Sales forecasting

---

55. Conclusion

The **APEX Retail Data Engineering Pipeline** successfully demonstrates the complete transformation of raw retail data into a trusted analytical platform.

The project starts with raw CSV data and progresses through:

Raw Data

↓

Data Ingestion

↓

Data Validation

↓

Data Cleaning

↓

Parquet

↓

Incremental Processing

↓

Delta Lake

↓

MERGE

↓

SCD Type 1 / Type 2

↓

Surrogate Keys

↓

Gold Star Schema

↓

Unity Catalog

↓

KPI Reporting

↓

RFM Analysis

↓

Cohort Retention

↓

Reconciliation

↓

Final Audit

The final system provides reliable and structured data for business analytics.

The implementation demonstrates practical knowledge of modern data engineering technologies and concepts, including Databricks, PySpark, Delta Lake, SQL, Unity Catalog, dimensional modeling, SCD, data quality, reconciliation, and business analytics.

---

56. Final Project Status

==

APEX RETAIL

PROJECT COMPLETION REPORT

==

DATA INGESTION

Raw / Inbound Processing : COMPLETED

CSV Processing : COMPLETED

Parquet Conversion : COMPLETED

DATA QUALITY

Customer Data Quality : PASS

Product Data Quality : PASS

Sales Data Quality : PASS

NULL Validation : PASS

Duplicate Validation : PASS

DATA ENGINEERING

Incremental Processing : COMPLETED

Delta Lake : COMPLETED

Delta MERGE : COMPLETED

SCD Type 1 : COMPLETED

SCD Type 2 : COMPLETED

Surrogate Keys : COMPLETED

DATA MODELING

Gold Star Schema : COMPLETED

dim_customer : COMPLETED

dim_product : COMPLETED

dim_promotion : COMPLETED

dim_date : COMPLETED

fact_sales : COMPLETED

DATA GOVERNANCE

Unity Catalog : COMPLETED

ANALYTICS

Sales Analytics : COMPLETED

Customer Analytics : COMPLETED

Product Analytics : COMPLETED

KPI Reporting : COMPLETED

RFM Analysis : COMPLETED

Cohort Retention : COMPLETED

VALIDATION

Customer Reconciliation : PASS

Product Reconciliation : PASS

Sales Reconciliation : PASS

Foreign-Key Validation : PASS

Transaction Validation : PASS

Gold Table Structure : PASS

Final Data Quality : PASS

==

FINAL PIPELINE STATUS: PASS

==

APEX RETAIL PIPELINE COMPLETED SUCCESSFULLY

==

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57. Appendix A — Important Unity Catalog Commands

Show Catalogs

SHOW CATALOGS;

Show Schemas

```sql
SHOW SCHEMAS IN apex_retail;
```

Show Gold Tables

```sql
SHOW TABLES IN apex_retail.gold;
```

View Fact Table

```sql
SELECT *
FROM apex_retail.gold.fact_sales
LIMIT 20;
```

Describe Table

```sql
DESCRIBE TABLE apex_retail.gold.fact_sales;
```

Describe Delta Table

```sql
DESCRIBE DETAIL apex_retail.gold.fact_sales;
```

---

58. Appendix B — Important Validation Results

Fact Foreign Keys

```
NULL Customer SKs  : 0
NULL Product SKs   : 0
NULL Promotion SKs : 0
NULL Date SKs      : 0
```

Transaction Integrity

```
NULL Transaction IDs      : 0
Duplicate Transaction IDs : 0
```

Reconciliation

```
Sales Reconciliation    : PASS
Customer Reconciliation : PASS
Product Reconciliation  : PASS
```

---

59. Appendix C — Evidence to Include in Word Report

The following screenshots should be inserted into the final Word documentation.

Screenshot — Raw Data Ingestion

Show:

- Input data
- DataFrame creation
- Record count

Screenshot — Data Quality

Show:

- NULL checks
- Duplicate checks
- PASS results

Screenshot — Delta Processing

Show:

- Delta table creation
- Delta format
- MERGE execution

Screenshot — SCD Processing

Show:

- SCD Type 1
- SCD Type 2
- Historical record handling

Screenshot 5 — Gold Star Schema

Show:

- Gold tables
- Dimension tables
- Fact table

Screenshot — Unity Catalog

Show:

```
apex_retail
    └── gold
        ├── dim_customer
        ├── dim_product
        ├── dim_promotion
        ├── dim_date
        └── fact_sales
```

Screenshot — KPI Reporting

Show:

- Revenue
- Transactions
- Quantity
- Other KPIs

Screenshot — RFM Analysis

Show:

- Recency
- Frequency
- Monetary
- Customer segments

Screenshot — Cohort Retention

Show:

- Cohort month
- Retention matrix
- Retention percentage

Screenshot — Final Audit

The most important screenshot should show:

FINAL PIPELINE STATUS: PASS

APEX RETAIL PIPELINE COMPLETED SUCCESSFULLY

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60. Final Statement

The APEX Retail project demonstrates the complete lifecycle of a modern data engineering solution, beginning with raw source data and ending with validated business-ready analytical data.

The project successfully combines data engineering, data quality, dimensional modeling, Delta Lake processing, Unity Catalog, and business analytics into a single end-to-end pipeline.

The resulting Gold layer provides a strong foundation for reporting, business intelligence, customer analytics, and future machine-learning applications.

APEX RETAIL DATA ENGINEERING PIPELINE — COMPLETED SUCCESSFULLY